

Experiment 3: Parallel Min, Max, Sum and Average using OpenMP Reduction

This guide explains how to perform Min, Max, Sum and Average operations using OpenMP reduction. The guide includes corrected code, execution steps for Windows and Linux, sample input/output, and viva questions.

Problem Statement: Implement Min, Max, Sum and Average operations using Parallel Reduction in OpenMP.

Correct Program Code

```
#include <iostream>
#include <omp.h>

using namespace std;

int main()
{
    int n;

    cout << "Enter number of elements: ";
    cin >> n;

    int* arr = new int[n];

    cout << "Enter elements:\n";

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    int minimum = arr[0];
    int maximum = arr[0];
    int sum = 0;

    double start_time = omp_get_wtime();

    // Parallel Min
    #pragma omp parallel for reduction(min:minimum)
    for (int i = 0; i < n; i++)
    {
        if (arr[i] < minimum)
        {
            minimum = arr[i];
        }
    }

    // Parallel Max
    #pragma omp parallel for reduction(max:maximum)
    for (int i = 0; i < n; i++)
    {
        if (arr[i] > maximum)
        {
            maximum = arr[i];
        }
    }

    // Parallel Sum
    #pragma omp parallel for reduction(+:sum)
    for (int i = 0; i < n; i++)
    {
        sum += arr[i];
    }

    double average = (double)sum / n;

    double end_time = omp_get_wtime();

    cout << "\nMinimum Value: " << minimum;
```

```

        cout << "\nMaximum Value: " << maximum;
        cout << "\nSum: " << sum;
        cout << "\nAverage: " << average;

        cout << "\n\nExecution Time: "
              << (end_time - start_time) * 1000
              << " milliseconds\n";

        delete[] arr;

        return 0;
    }

```

How to Run on Windows

1. Install Code::Blocks with MinGW compiler.
2. Open Code::Blocks.
3. Go to File → New → Empty File.
4. Paste the program code.
5. Save file as reduction_operations.cpp.
6. Go to Settings → Compiler.
7. Open Other Compiler Options.
8. Add: -fopenmp
9. Click OK.
10. Press F9 or click Build and Run.

How to Run on Linux / Ubuntu

Install required software:

```

sudo apt update
sudo apt install g++
sudo apt install libomp-dev

```

Compile and run:

```

g++ -fopenmp reduction_operations.cpp -o reduction_operations

./reduction_operations

```

Sample Input

```

Enter number of elements: 5

Enter elements:
4
2
7
1
9

```

Expected Output

```

Minimum Value: 1
Maximum Value: 9
Sum: 23
Average: 4.6

Execution Time: 20 milliseconds

```

Important Viva Questions

Question	Answer
What is reduction?	Reduction combines thread results into one final variable.
Which reduction operators are used?	min, max and +
Why use OpenMP?	To perform operations in parallel using multiple CPU threads.
What is advantage of reduction?	Reduction avoids race conditions.